

Fig. 12: Message being transmitted three times at 50ms intervals

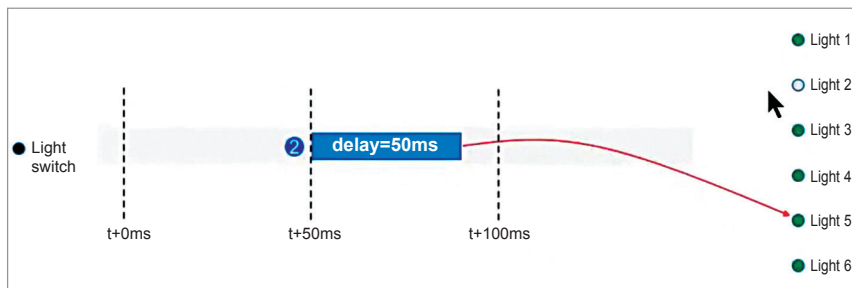


Fig. 13: Second copy is sent 50ms later and is received by the 5th of the six lights

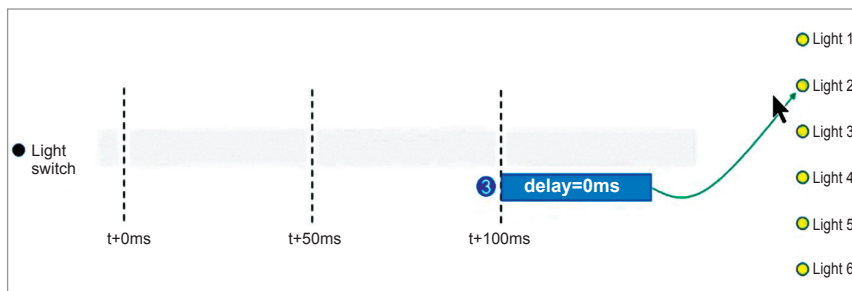


Fig. 14: Final copy has no delay specified

Some key issues about Bluetooth mesh

Advertising bearer. The main bearer used in Bluetooth mesh is connectionless and, as such, there is no mechanism for coordinating with the scanning and advertising schedules of other nodes in the network.

Rx duty cycle. To maximise the probability of messages being received, the Rx duty cycle should be high. However, a RX duty cycle of 100% is rarely achievable.

Acknowledgements. Acknowledged messages can be useful in one-to-one communication but do not scale in multicast messaging to groups of arbitrary sizes; so unacknowledged messages should normally be used.

Stochastic system. Includes inherent randomness and unpredictability.

capacity to receive packets. If that is zero, then the receiver's buffer is full and the transmitter will wait before resuming packet transmission again.

So, it is a dynamic flow control strategy, which prevents buffer overflow.

Bluetooth LE reliability for connectionless communication

In Bluetooth, connectionless communication is called advertising. In the

basic case, copies of each packet are broadcast on each channel in random channel order. Their time intervals are solely controlled by the broadcasting device, without any coordination with the transmitter. There is also another form of advertising called extended advertising, which uses all 40 Bluetooth LE channels.

The use of three channels in basic advertising is less sophisticated than adaptive frequency hopping in a con-

nection. But the advertising channels are deliberately placed far apart from each other so that interference in one part does not block communication completely.

Basic advertising happens at defined intervals, but it includes some random timing variation in the scheduling of packet transmission of up to ten milliseconds. This is to avoid persistent collisions with nearby devices that happen to be transmitting or advertising on the same schedule.

Bluetooth connectionless communication does not attempt to coordinate the activities of the advertising device with that of the receiving devices. However, there is a special advertising mode called periodic advertising, which makes the timing of the advertising completely deterministic and precise.

And scanning devices can also discover the periodic advertising schedule of the broadcasting device so that they can synchronise their scanning activity precisely with it. This way, devices can more reliably and efficiently receive broadcast data than with the basic form of advertising.

Bluetooth mesh networks

Bluetooth mesh allows creation of large networks, having tens and thousands of devices. It is a special way of using Bluetooth LE where the mesh stack sits on top of the Bluetooth LE controller.

Bluetooth mesh defines bearers, which provide different ways of using the Bluetooth LE controller for the communication of mesh messages. And the bearers provide support for both connectionless and connection-oriented communication.

Addressing Bluetooth mesh issues

Mesh nodes can be configured to automatically retransmit messages at defined intervals, so that multiple copies get rapidly retransmitted. And, consequently, the probability of message loss drops dramatically when you send multiple copies. If you want to ensure that messages are acted upon by all receiving devices at the same time (even if they do not receive the same copy), that can be done too by